

# CohortCharacteristics

An R package to characterise cohorts

2026-09-17

# CohortCharacteristics



# Website

CohortCharacteristics is on CRAN:

```
install.packages("CohortCharacteristics")
```

The package documentation and vignettes are available on the website:

<https://darwin-eu.github.io/CohortCharacteristics/>

# Set up

```
options("width"=115)
library(duckdb)
library(CDMConnector)
library(PatientProfiles)
library(CohortConstructor)
library(dplyr)
library(here)
library(CohortCharacteristics)

cdm <- omock::mockCdmFromDataset(datasetName = "GiBleed",
                                source = "duckdb")
```

# Let's get started

Let's create some cohorts that we will use during the presentation:

```
codes <- c(list(anemia = 439777,
               rheumatoid_arthritis = 80809,
               osteoarthritis = 80180,
               pneumonia = 255848))

cdm[["my_cohort"]] <- conceptCohort(cdm = cdm,
                                   conceptSet = codes,
                                   name = "my_cohort")

cdm[["my_cohort"]] |>
  glimpse()
```

Rows: ??

Columns: 4

\$ cohort\_definition\_id <int> 2, 2, 2, 2, 2, 2, 2, 2, 1, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, ...

\$ subject\_id <int> 171, 1083, 1154, 1478, 1695, 1744, 4027, 4117, 1497, 61, 1432, 2358, 2597, 3298, 348...

\$ cohort\_start\_date <date> 2007-12-21, 2005-04-29, 2006-07-14, 2001-12-26, 2010-05-03, 2010-08-06, 2008-04-16, ...

\$ cohort\_end\_date <date> 2007-12-21, 2005-04-29, 2006-07-14, 2001-12-26, 2010-05-03, 2010-08-06, 2008-04-16, ...

# Workflow

We have three types of functions:

- **summarise**: these functions produce a standardised summary of a cohort. This output is called `<summarised_result>`.
- **table**: these functions produce tables from the `<summarised_result>` object generated by the *summarise* functions.
- **plot**: these functions produce plots from the `<summarised_result>` object generated by the *summarise* functions. (We won't show these today)



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# summariseCohortCount

Use `summariseCohortCount()` to get counts for your cohort. Pipe your cohort table into the function as shown below.

```
results <- cdm$my_cohort |>
  summariseCohortCount()
```

It returns cohort counts in a `summarised_result` object.

```
glimpse(results)
```

```
Rows: 8
Columns: 13
$ result_id      <int> 1, 1, 1, 1, 1, 1, 1, 1
$ cdm_name       <chr> "GiBleed", "GiBleed", "GiBleed", "GiBleed", "GiBleed", "GiBleed", "GiBleed", "GiBleed"
$ group_name     <chr> "cohort_name", "cohort_name", "cohort_name", "cohort_name", "cohort_name", "cohort_name", "cohort_name"...
$ group_level    <chr> "anemia", "anemia", "osteoarthritis", "osteoarthritis", "pneumonia", "pneumonia", "rheum..."
$ strata_name    <chr> "overall", "overall", "overall", "overall", "overall", "overall", "overall", "overall"
$ strata_level   <chr> "overall", "overall", "overall", "overall", "overall", "overall", "overall", "overall"
$ variable_name  <chr> "Number records", "Number subjects", "Number records", "Number subjects", "Number record..."
$ variable_level <chr> NA, NA, NA, NA, NA, NA, NA, NA
$ estimate_name  <chr> "count", "count", "count", "count", "count", "count", "count", "count"
$ estimate_type  <chr> "integer", "integer", "integer", "integer", "integer", "integer", "integer", "integer"
$ estimate_value <chr> "100", "100", "2630", "2630", "52", "52", "18", "18"
$ additional_name <chr> "overall", "overall", "overall", "overall", "overall", "overall", "overall", "overall"
```



# summariseCohortCount

## Note

Previously we saw the function `cohortCount()` to see the counts of a `<cohort_table>`.

```
cohortCount(cdm$my_cohort)
```

```
# A tibble: 4 × 3
  cohort_definition_id number_records number_subjects
      <int>           <int>           <int>
1             1           100             100
2             2          2630            2630
3             3            52             52
4             4            18             18
```

The function `summariseCohortCount()` produces a similar result but in a standardised output (`<summarised_result>`).

# tableCohortCount

To get a tidy table from the `<summarised_result>` object.

```
tableCohortCount(result = results)
```

| CDM name | Variable name   | Estimate name | Cohort name |                |           |                      |
|----------|-----------------|---------------|-------------|----------------|-----------|----------------------|
|          |                 |               | anemia      | osteoarthritis | pneumonia | rheumatoid_arthritis |
| GiBleed  | Number records  | N             | 100         | 2,630          | 52        | 18                   |
|          | Number subjects | N             | 100         | 2,630          | 52        | 18                   |

# summariseCohortAttrition

To summarise cohort attrition, pipe your cohort table into the function as shown below.

```
results <- cdm$my_cohort |>
  summariseCohortAttrition()
```

It returns cohort attrition in a **summarised\_result** object.

```
glimpse(results)
```

```

Rows: 64
Columns: 13
$ result_id      <int> 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2...
$ cdm_name       <chr> "GiBleed", "GiBleed", "GiBleed", "GiBleed", "GiBleed", "GiBleed", "GiBleed", "GiBleed", "GiBleed", ...
$ group_name     <chr> "cohort_name", "cohort_name", "cohort_name", "cohort_name", "cohort_name", "cohort_name", "cohort_name"...
$ group_level    <chr> "anemia", "anemia", "anemia", "anemia", "anemia", "anemia", "anemia", "anemia", "anemia", "anemia"...
$ strata_name    <chr> "reason", "reason", "reason", "reason", "reason", "reason", "reason", "reason", "reason", "reason"...
$ strata_level   <chr> "Initial qualifying events", "Initial qualifying events", "Initial qualifying events", "...
$ variable_name  <chr> "number_records", "number_subjects", "excluded_records", "excluded_subjects", "number_re...
$ variable_level <chr> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, ...
$ estimate_name  <chr> "count", "count", "count", "count", "count", "count", "count", "count", "count", "count"...
$ estimate_type  <chr> "integer", "integer", "integer", "integer", "integer", "integer", "integer", "integer", "integer", ...
$ estimate_value <chr> "102", "102", "0", "0", "100", "100", "2", "2", "100", "100", "0", "0", "100", "100", "0...
$ additional_name <chr> "reason_id", "reason_id", "reason_id", "reason_id", "reason_id", "reason_id", "reason_id", "reason_id"...

```

# summariseCohortAttrition



## Note

Yesterday we saw the function `attrition()` to see the attrition of a `<cohort_table>`.

```
attrition(cdm$my_cohort)
```

```
# A tibble: 16 × 7
```

|    | cohort_definition_id | number_records | number_subjects | reason_id | reason            | excluded_records | excluded_subjects |
|----|----------------------|----------------|-----------------|-----------|-------------------|------------------|-------------------|
|    | <int>                | <int>          | <int>           | <int>     | <chr>             | <int>            | <int>             |
| 1  | 1                    | 102            | 102             | 1         | Initial qualif... | 0                | 0                 |
| 2  | 1                    | 100            | 100             | 2         | Record in obse... | 2                | 2                 |
| 3  | 1                    | 100            | 100             | 3         | Not missing re... | 0                | 0                 |
| 4  | 1                    | 100            | 100             | 4         | Merge overlapp... | 0                | 0                 |
| 5  | 2                    | 2694           | 2694            | 1         | Initial qualif... | 0                | 0                 |
| 6  | 2                    | 2630           | 2630            | 2         | Record in obse... | 64               | 64                |
| 7  | 2                    | 2630           | 2630            | 3         | Not missing re... | 0                | 0                 |
| 8  | 2                    | 2630           | 2630            | 4         | Merge overlapp... | 0                | 0                 |
| 9  | 3                    | 52             | 52              | 1         | Initial qualif... | 0                | 0                 |
| 10 | 3                    | 52             | 52              | 2         | Record in obse... | 0                | 0                 |
| 11 | 3                    | 52             | 52              | 3         | Not missing re... | 0                | 0                 |
| 12 | 3                    | 52             | 52              | 4         | Merge overlapp... | 0                | 0                 |
| 13 | 4                    | 18             | 18              | 1         | Initial qualif... | 0                | 0                 |
| 14 | 4                    | 18             | 18              | 2         | Record in obse... | 0                | 0                 |
| 15 | 4                    | 18             | 18              | 3         | Not missing re... | 0                | 0                 |
| 16 | 4                    | 18             | 18              | 4         | Merge overlapp... | 0                | 0                 |

The function `summariseCohortAttrition()` produces a similar result but in a standardised output (`<summarised_result>`).

# tableCohortAttrition

To get a tidy table from the `<summarised_result>` object.

```
tableCohortAttrition(result = results)
```

| Reason                        | Variable name  |                 |                  |                   |
|-------------------------------|----------------|-----------------|------------------|-------------------|
|                               | number_records | number_subjects | excluded_records | excluded_subjects |
| GiBleed; anemia               |                |                 |                  |                   |
| Initial qualifying events     | 102            | 102             | 0                | 0                 |
| Record in observation         | 100            | 100             | 2                | 2                 |
| Not missing record date       | 100            | 100             | 0                | 0                 |
| Merge overlapping records     | 100            | 100             | 0                | 0                 |
| GiBleed; osteoarthritis       |                |                 |                  |                   |
| Initial qualifying events     | 2,694          | 2,694           | 0                | 0                 |
| Record in observation         | 2,630          | 2,630           | 64               | 64                |
| Not missing record date       | 2,630          | 2,630           | 0                | 0                 |
| Merge overlapping records     | 2,630          | 2,630           | 0                | 0                 |
| GiBleed; pneumonia            |                |                 |                  |                   |
| Initial qualifying events     | 52             | 52              | 0                | 0                 |
| Record in observation         | 52             | 52              | 0                | 0                 |
| Not missing record date       | 52             | 52              | 0                | 0                 |
| Merge overlapping records     | 52             | 52              | 0                | 0                 |
| GiBleed; rheumatoid_arthritis |                |                 |                  |                   |
| Initial qualifying events     | 18             | 18              | 0                | 0                 |
| Record in observation         | 18             | 18              | 0                | 0                 |

# summariseCharacteristics

The main functionality of the package is to extract cohort characteristics with `summariseCharacteristics()`.

```
results <- cdm$my_cohort |>
  summariseCharacteristics()
```

It returns cohort characteristics in a `<summarised_result>` object.

```
glimpse(results)
```

Rows: 204

Columns: 13

```
$ result_id      <int> 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1...
$ cdm_name      <chr> "GiBleed", "GiBleed", "GiBleed", "GiBleed", "GiBleed", "GiBleed", "GiBleed", "GiBleed", "GiBleed", ...
$ group_name    <chr> "cohort_name", "cohort_name", "cohort_name", "cohort_name", "cohort_name", "cohort_name", "cohort_name"...
$ group_level   <chr> "anemia", "anemia", "anemia", "anemia", "anemia", "anemia", "anemia", "anemia", "anemia", "anemia"...
$ strata_name   <chr> "overall", "overall", "overall", "overall", "overall", "overall", "overall", "overall", "overall", ...
$ strata_level  <chr> "overall", "overall", "overall", "overall", "overall", "overall", "overall", "overall", "overall", ...
$ variable_name <chr> "Number records", "Number subjects", "Cohort start date", "Cohort start date", "Cohort s...
$ variable_level <chr> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, "Female", "F...
$ estimate_name <chr> "count", "count", "min", "q25", "median", "q75", "max", "min", "q25", "median", "q75", "...
$ estimate_type <chr> "integer", "integer", "date", "date", "date", "date", "date", "date", "date", "date", "date", "d...
$ estimate_value <chr> "100", "100", "1912-04-30", "1974-01-30", "1990-12-04", "2005-08-09", "2018-11-08", "191...
$ additional_name <chr> "overall", "overall", "overall", "overall", "overall", "overall", "overall", "overall", "overall", ...
```



# tableCharacteristics

To get a tidy table from the `<summarised_result>` object.

```
tableCharacteristics(result = results, header = c("cdm_name", "cohort_name"))
```

|                    |                |                    | CDM name                             |                                      |                                      |                                      |
|--------------------|----------------|--------------------|--------------------------------------|--------------------------------------|--------------------------------------|--------------------------------------|
|                    |                |                    | GiBleed                              |                                      |                                      |                                      |
|                    |                |                    | Cohort name                          |                                      |                                      |                                      |
| Variable name      | Variable level | Estimate name      | anemia                               | osteoarthritis                       | pneumonia                            | rheumatoid_arthritis                 |
| Number records     | –              | N                  | 100                                  | 2,630                                | 52                                   | 18                                   |
| Number subjects    | –              | N                  | 100                                  | 2,630                                | 52                                   | 18                                   |
| Cohort start date  | –              | Median [Q25 - Q75] | 1990-12-04 [1974-01-30 - 2005-08-09] | 1999-11-16 [1988-03-03 - 2008-08-07] | 2007-07-06 [1998-09-17 - 2013-04-07] | 2000-08-10 [1992-01-18 - 2008-01-01] |
|                    |                | Range              | 1912-04-30 to 2018-11-08             | 1941-12-05 to 2019-04-14             | 1980-03-03 to 2019-02-18             | 1971-07-29 to 2017-06-06             |
| Cohort end date    | –              | Median [Q25 - Q75] | 1990-12-04 [1974-01-30 - 2005-08-09] | 1999-11-16 [1988-03-03 - 2008-08-07] | 2007-07-06 [1998-09-17 - 2013-04-07] | 2004-11-17 [1993-03-05 - 2011-05-22] |
|                    |                | Range              | 1912-04-30 to 2018-11-08             | 1941-12-05 to 2019-04-14             | 1980-03-03 to 2019-02-18             | 1971-07-29 to 2017-06-06             |
| Age                | –              | Median [Q25 - Q75] | 32.00 [17.75 - 50.25]                | 38.00 [36.00 - 41.00]                | 82.00 [78.00 - 86.25]                | 41.00 [34.25 - 55.00]                |
|                    |                | Mean (SD)          | 33.49 (22.14)                        | 38.15 (3.27)                         | 80.33 (9.48)                         | 45.22 (15.11)                        |
|                    |                | Range              | 0.00 to 81.00                        | 31.00 to 46.00                       | 53.00 to 96.00                       | 24.00 to 77.00                       |
| Sex                | Female         | N (%)              | 78 (78.00%)                          | 1,341 (50.99%)                       | 36 (69.23%)                          | 8 (44.44%)                           |
|                    | Male           | N (%)              | 22 (22.00%)                          | 1,289 (49.01%)                       | 16 (30.77%)                          | 10 (55.56%)                          |
| Prior observation  | –              | Median [Q25 - Q75] | 12,045.00 [6,843.00 - 18,706.00]     | 14,097.00 [13,161.25 - 15,039.00]    | 30,079.00 [28,538.25 - 31,726.50]    | 15,330.00 [12,866.25 - 20,440.00]    |
|                    |                | Mean (SD)          | 12,570.42 (8,107.27)                 | 14,105.97 (1,193.14)                 | 29,498.13 (3,467.33)                 | 16,870.94 (5,513.54)                 |
|                    |                | Range              | 0.00 to 29,930.00                    | 11,340.00 to 17,044.00               | 19,485.00 to 35,210.00               | 9,125.00 to 28,469.00                |
| Future observation | –              | Median [Q25 - Q75] | 9,641.50 [4,596.00 - 14,747.00]      | 6,790.00 [3,568.00 - 10,012.00]      | 12,000 [10,000 - 15,000]             | 6,065.50 [4,058.00 - 9,716.00]       |

# Commonly used arguments for summariseCharacteristics

- **strata**: creates stratified results.
- **cohortId**: filters by cohort ID.
- **ageGroup**: defines the age groups to summarise.
- **tableIntersect**: count/flag/date/days
- **cohortIntersect**: count/flag/date/days
- **conceptIntersect**: count/flag/date/days
- **link to documentation**



# ageGroup example

```
result <- summariseCharacteristics(cdm$my_cohort, ageGroup = list(c(0,10), c(11,18), c(19,150)))  
tableCharacteristics(result = result, header = "cohort_name")
```

# ageGroup example

| CDM name | Variable name      | Variable level | Estimate name      | Cohort name                          |                                      |                                      |                                      |
|----------|--------------------|----------------|--------------------|--------------------------------------|--------------------------------------|--------------------------------------|--------------------------------------|
|          |                    |                |                    | anemia                               | osteoarthritis                       | pneumonia                            | rheumatoid_arthritis                 |
| GiBleed  | Number records     | —              | N                  | 100                                  | 2,630                                | 52                                   | 18                                   |
|          | Number subjects    | —              | N                  | 100                                  | 2,630                                | 52                                   | 18                                   |
|          | Cohort start date  | —              | Median [Q25 - Q75] | 1990-12-04 [1974-01-30 - 2005-08-09] | 1999-11-16 [1988-03-03 - 2008-08-07] | 2007-07-06 [1998-09-17 - 2013-04-07] | 2000-08-10 [1992-01-18 - 2008-01-01] |
|          |                    |                | Range              | 1912-04-30 to 2018-11-08             | 1941-12-05 to 2019-04-14             | 1980-03-03 to 2019-02-18             | 1971-07-29 to 2017-06-06             |
|          | Cohort end date    | —              | Median [Q25 - Q75] | 1990-12-04 [1974-01-30 - 2005-08-09] | 1999-11-16 [1988-03-03 - 2008-08-07] | 2007-07-06 [1998-09-17 - 2013-04-07] | 2004-11-17 [1993-03-05 - 2011-05-22] |
|          |                    |                | Range              | 1912-04-30 to 2018-11-08             | 1941-12-05 to 2019-04-14             | 1980-03-03 to 2019-02-18             | 1971-07-29 to 2017-06-06             |
|          | Age                | —              | Median [Q25 - Q75] | 32.00 [17.75 - 50.25]                | 38.00 [36.00 - 41.00]                | 82.00 [78.00 - 86.25]                | 41.00 [34.25 - 55.00]                |
|          |                    |                | Mean (SD)          | 33.49 (22.14)                        | 38.15 (3.27)                         | 80.33 (9.48)                         | 45.22 (15.11)                        |
|          |                    |                | Range              | 0.00 to 81.00                        | 31.00 to 46.00                       | 53.00 to 96.00                       | 24.00 to 77.00                       |
|          | Age group          | 0 to 10        | N (%)              | 20 (20.00%)                          | 0 (0.00%)                            | 0 (0.00%)                            | 0 (0.00%)                            |
|          |                    | 11 to 18       | N (%)              | 7 (7.00%)                            | 0 (0.00%)                            | 0 (0.00%)                            | 0 (0.00%)                            |
|          |                    | 19 to 150      | N (%)              | 73 (73.00%)                          | 2,630 (100.00%)                      | 52 (100.00%)                         | 18 (100.00%)                         |
|          | Sex                | Female         | N (%)              | 78 (78.00%)                          | 1,341 (50.99%)                       | 36 (69.23%)                          | 8 (44.44%)                           |
|          |                    | Male           | N (%)              | 22 (22.00%)                          | 1,289 (49.01%)                       | 16 (30.77%)                          | 10 (55.56%)                          |
|          | Prior observation  | —              | Median [Q25 - Q75] | 12,045.00 [6,843.00 - 18,706.00]     | 14,097.00 [13,161.25 - 15,039.00]    | 30,079.00 [28,538.25 - 31,726.50]    | 15,330.00 [12,866.25 - 20,440.00]    |
|          |                    |                | Mean (SD)          | 12,570.42 (8,107.27)                 | 14,105.97 (1,193.14)                 | 29,498.13 (3,467.33)                 | 16,870.94 (5,513.54)                 |
|          |                    |                | Range              | 0.00 to 29,930.00                    | 11,340.00 to 17,044.00               | 19,485.00 to 35,210.00               | 9,125.00 to 28,469.00                |
|          | Future observation |                | Median [Q25 - Q75] | 9,641.50 [4,596.00 -                 | 6,790.00 [3,568.00 -                 | 12,000.00 [10,000 - 15,000]          | 6,065.50 [4,058.00 -                 |

# Strata example

```
result <- cdm$my_cohort |>
  addAge(ageGroup = list(c(0,10),c(11,18), c(19,150))) |>
  summariseCharacteristics(cohortId = "anemia",
                          strata = list("age_group"))
tableCharacteristics(result = result, header = c("cdm_name", "cohort_name","age_group"))
```

# Strata example

|                   |                |                    | CDM name                             |                                      |                                      |                                      |
|-------------------|----------------|--------------------|--------------------------------------|--------------------------------------|--------------------------------------|--------------------------------------|
|                   |                |                    | GiBleed                              |                                      |                                      |                                      |
|                   |                |                    | Cohort name                          |                                      |                                      |                                      |
|                   |                |                    | anemia                               |                                      |                                      |                                      |
| Variable name     | Variable level | Estimate name      | Age group                            |                                      |                                      |                                      |
|                   |                |                    | overall                              | 0 to 10                              | 11 to 18                             | 19 to 150                            |
| Number records    | –              | N                  | 100                                  | 20                                   | 7                                    | 73                                   |
| Number subjects   | –              | N                  | 100                                  | 20                                   | 7                                    | 73                                   |
| Cohort start date | –              | Median [Q25 - Q75] | 1990-12-04 [1974-01-30 - 2005-08-09] | 1963-03-26 [1956-03-03 - 1972-12-27] | 1969-08-19 [1963-03-03 - 1975-04-11] | 1999-03-30 [1987-11-03 - 2007-06-08] |
|                   |                | Range              | 1912-04-30 to 2018-11-08             | 1912-04-30 to 1987-01-08             | 1951-04-05 to 1986-04-07             | 1939-01-29 to 2018-11-08             |
| Cohort end date   | –              | Median [Q25 - Q75] | 1990-12-04 [1974-01-30 - 2005-08-09] | 1963-03-26 [1956-03-03 - 1972-12-27] | 1969-08-19 [1963-03-03 - 1975-04-11] | 1999-03-30 [1987-11-03 - 2007-06-08] |
|                   |                | Range              | 1912-04-30 to 2018-11-08             | 1912-04-30 to 1987-01-08             | 1951-04-05 to 1986-04-07             | 1939-01-29 to 2018-11-08             |
| Age               | –              | Median [Q25 - Q75] | 32.00 [17.75 - 50.25]                | 2.50 [0.75 - 5.25]                   | 17.00 [14.50 - 17.50]                | 43.00 [28.00 - 56.00]                |
|                   |                | Mean (SD)          | 33.49 (22.14)                        | 3.30 (3.15)                          | 15.86 (2.27)                         | 43.45 (16.93)                        |
|                   |                | Range              | 0.00 to 81.00                        | 0.00 to 9.00                         | 12.00 to 18.00                       | 19.00 to 81.00                       |
| Sex               | Female         | N (%)              | 78 (78.00%)                          | 10 (50.00%)                          | 6 (85.71%)                           | 62 (84.93%)                          |
|                   | Male           | N (%)              | 22 (22.00%)                          | 10 (50.00%)                          | 1 (14.29%)                           | 11 (15.07%)                          |
| Prior observation | –              | Median [Q25 - Q75] | 12,045.00 [6,843.00 - 18,706.00]     | 1,094.50 [365.00 - 2,280.25]         | 6,570.00 [5,657.00 - 6,752.00]       | 16,060.00 [10,585.00 - 20,804.00]    |
|                   |                | Mean (SD)          | 12,570.42 (8,107.27)                 | 1,478.00 (1,221.91)                  | 6,152.57 (827.72)                    | 16,224.85 (6,180.75)                 |
|                   |                | Range              | 0.00 to 29,930.00                    | 0.00 to 3,650.00                     | 4,745.00 to 6,935.00                 | 7,300.00 to 29,930.00                |
|                   |                |                    |                                      |                                      |                                      |                                      |

# Intersection with another table

You can get count/flag/date/days summaries with other tables.

```
result <- cdm$my_cohort |>
  summariseCharacteristics(
    cohortId = "anemia",
    tableIntersectCount = list(
      "Number of visits prior year" = list(
        tableName = "visit_occurrence",
        window = c(-365, 0)
      )
    ),
    cohortIntersectFlag = list(
      "Previous rheumatoid arthritis diagnosis" = list(
        targetCohortTable = "my_cohort",
        targetCohortId = "rheumatoid_arthritis",
        window = c(-Inf, 0)
      )
    ),
    conceptIntersectDays = list(
      "Time since last X-ray" = list(
```

# Intersection with another table

You can get count/flag/date/days summaries with other tables.

| CDM name | Variable name      | Variable level | Estimate name      | Cohort name                          |                                      |                                      |                                      |
|----------|--------------------|----------------|--------------------|--------------------------------------|--------------------------------------|--------------------------------------|--------------------------------------|
|          |                    |                |                    | anemia                               | osteoarthritis                       | pneumonia                            | rheumatoid_arthritis                 |
| GiBleed  | Number records     | —              | N                  | 100                                  | 2,630                                | 52                                   | 18                                   |
|          | Number subjects    | —              | N                  | 100                                  | 2,630                                | 52                                   | 18                                   |
|          | Cohort start date  | —              | Median [Q25 - Q75] | 1990-12-04 [1974-01-30 - 2005-08-09] | 1999-11-16 [1988-03-03 - 2008-08-07] | 2007-07-06 [1998-09-17 - 2013-04-07] | 2000-08-10 [1992-01-18 - 2008-01-01] |
|          |                    |                | Range              | 1912-04-30 to 2018-11-08             | 1941-12-05 to 2019-04-14             | 1980-03-03 to 2019-02-18             | 1971-07-29 to 2017-06-06             |
|          | Cohort end date    | —              | Median [Q25 - Q75] | 1990-12-04 [1974-01-30 - 2005-08-09] | 1999-11-16 [1988-03-03 - 2008-08-07] | 2007-07-06 [1998-09-17 - 2013-04-07] | 2004-11-17 [1993-03-05 - 2011-05-22] |
|          |                    |                | Range              | 1912-04-30 to 2018-11-08             | 1941-12-05 to 2019-04-14             | 1980-03-03 to 2019-02-18             | 1971-07-29 to 2017-06-06             |
|          | Age                | —              | Median [Q25 - Q75] | 32.00 [17.75 - 50.25]                | 38.00 [36.00 - 41.00]                | 82.00 [78.00 - 86.25]                | 41.00 [34.25 - 55.00]                |
|          |                    |                | Mean (SD)          | 33.49 (22.14)                        | 38.15 (3.27)                         | 80.33 (9.48)                         | 45.22 (15.11)                        |
|          |                    |                | Range              | 0.00 to 81.00                        | 31.00 to 46.00                       | 53.00 to 96.00                       | 24.00 to 77.00                       |
|          | Sex                | Female         | N (%)              | 78 (78.00%)                          | 1,341 (50.99%)                       | 36 (69.23%)                          | 8 (44.44%)                           |
|          |                    | Male           | N (%)              | 22 (22.00%)                          | 1,289 (49.01%)                       | 16 (30.77%)                          | 10 (55.56%)                          |
|          | Prior observation  | —              | Median [Q25 - Q75] | 12,045.00 [6,843.00 - 18,706.00]     | 14,097.00 [13,161.25 - 15,039.00]    | 30,079.00 [28,538.25 - 31,726.50]    | 15,330.00 [12,866.25 - 20,440.00]    |
|          |                    |                | Mean (SD)          | 12,570.42 (8,107.27)                 | 14,105.97 (1,193.14)                 | 29,498.13 (3,467.33)                 | 16,870.94 (5,513.54)                 |
|          |                    |                | Range              | 0.00 to 29,930.00                    | 11,340.00 to 17,044.00               | 19,485.00 to 35,210.00               | 9,125.00 to 28,469.00                |
|          | Future observation | —              | Median [Q25 - Q75] | 9,641.50 [4,596.00 - 15,796.25]      | 6,790.00 [3,568.00 - 10,781.25]      | 12.00 [10.00 - 15.00]                | 6,065.50 [4,058.00 - 9,716.00]       |
|          |                    |                | Mean (SD)          | 10,773.47 (7,880.91)                 | 7,686.76 (5,364.39)                  | 12.25 (3.16)                         | 6,978.83 (4,735.79)                  |
|          |                    |                | Range              | 1.00 to 39,143.00                    | 10.00 to 28,328.00                   | 7.00 to 19.00                        | 668.00 to 17,356.00                  |
|          |                    |                |                    |                                      |                                      |                                      |                                      |



# Intersections

## Note

The arguments of the intersection functions match the *PatientProfiles* functions that we have just seen:

- `tableIntersectFlag -> PatientProfiles::addTableIntersectFlag()`
- `tableIntersectCount -> PatientProfiles::addTableIntersectCount()`
- `tableIntersectDays -> PatientProfiles::addTableIntersectDays()`
- `tableIntersectDate -> PatientProfiles::addTableIntersectDate()`
- `conceptIntersectFlag -> PatientProfiles::addConceptIntersectFlag()`
- `conceptIntersectCount -> PatientProfiles::addConceptIntersectCount()`
- `conceptIntersectDays -> PatientProfiles::addConceptIntersectDays()`
- `conceptIntersectDate -> PatientProfiles::addConceptIntersectDate()`
- `cohortIntersectFlag -> PatientProfiles::addCohortIntersectFlag()`
- `cohortIntersectCount -> PatientProfiles::addCohortIntersectCount()`
- `cohortIntersectDays -> PatientProfiles::addCohortIntersectDays()`
- `cohortIntersectDate -> PatientProfiles::addCohortIntersectDate()`



# Your turn!



# Exercise 1 - characterise an osteoarthritis cohort

```
cdm <- omock::mockCdmFromDataset(datasetName = "GiBleed",  
                                  source = "duckdb")  
  
cdm[["my_cohort"]] <- conceptCohort(cdm = cdm,  
                                   conceptSet = codes,  
                                   name = "my_cohort")
```

Can you characterise the **osteoarthritis** cohort?

- Start with only demographics
- Add if they had any anemia diagnosis during the last 180 days (including index date).
- Add if they had any pneumonia diagnosis last 2 years (excluding index date).

►  **Click to see solution**

# summariseLargeScaleCharacteristics

Sometimes we might want to summarise all clinical events for cohorts across different time windows. We can do this with `summariseLargeScaleCharacteristics()`.

```
result <- cdm$my_cohort |>
  summariseLargeScaleCharacteristics(
    window = list(c(-Inf, -1), c(1, Inf)),
    eventInWindow = "condition_occurrence",
    minimumFrequency = 0.05
  )
```

```
glimpse(result)
```

Rows: 250

Columns: 13

```
$ result_id      <int> 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1...
$ cdm_name       <chr> "GiBleed", "GiBleed", "GiBleed", "GiBleed", "GiBleed", "GiBleed", "GiBleed", "GiBleed", "GiBleed", ...
$ group_name     <chr> "cohort_name", "cohort_name", "cohort_name", "cohort_name", "cohort_name", "cohort_name", "cohort_name"...
$ group_level    <chr> "anemia", "anemia", "anemia", "anemia", "anemia", "anemia", "anemia", "anemia", "anemia", "anemia"...
$ strata_name    <chr> "overall", "overall", "overall", "overall", "overall", "overall", "overall", "overall", "overall", ...
$ strata_level   <chr> "overall", "overall", "overall", "overall", "overall", "overall", "overall", "overall", "overall", ...
$ variable_name  <chr> "Acute allergic reaction", "Acute allergic reaction", "Acute bronchitis", "Acute bronchi...
$ variable_level <chr> "-inf to -1", "-inf to -1", "-inf to -1", "-inf to -1", "-inf to -1", "-inf to -1", "-inf to -1", "-in...
$ estimate_name  <chr> "count", "percentage", "count", "percentage", "count", "percentage", "count", "percentage", "count", "percentage"...
$ estimate_type  <chr> "integer", "percentage", "integer", "percentage", "integer", "percentage", "integer", "percentage", "integer", "p...
$ estimate_value <chr> "5", "5.10", "77", "78.57", "75", "76.53", "7", "7.14", "6", "6.12", "6", "6.12", "7", "7...
$ additional_name <chr> "concept_id", "concept_id", "concept_id", "concept_id", "concept_id", "concept_id", "concept_id", "con..."
```

# tableTopLargeScaleCharacteristics

`tableTopLargeScaleCharacteristics()` creates a tidy table of the most frequent concepts.

```
tableTopLargeScaleCharacteristics(result = result, topConcepts = 5)
```

| Top | Cohort name                                     |                                                 |                                                   |                                                   |                                                 |                                                 |                                                 |
|-----|-------------------------------------------------|-------------------------------------------------|---------------------------------------------------|---------------------------------------------------|-------------------------------------------------|-------------------------------------------------|-------------------------------------------------|
|     | anemia                                          |                                                 | osteoarthritis                                    |                                                   | pneumonia                                       | rheumatoid_arthritis                            |                                                 |
|     |                                                 |                                                 | Window                                            |                                                   |                                                 |                                                 |                                                 |
|     | -inf to -1                                      | 1 to inf                                        | -inf to -1                                        | 1 to inf                                          | -inf to -1                                      | -inf to -1                                      | 1 to inf                                        |
| 1   | Viral sinusitis (40481087)<br>82 (83.7%)        | Viral sinusitis (40481087)<br>82 (82.0%)        | Viral sinusitis (40481087)<br>2594 (98.6%)        | Viral sinusitis (40481087)<br>2066 (78.6%)        | Alzheimer's disease (378419)<br>43 (82.7%)      | Acute viral pharyngitis (4112343)<br>17 (94.4%) | Viral sinusitis (40481087)<br>13 (72.2%)        |
| 2   | Acute bronchitis (260139)<br>77 (78.6%)         | Acute viral pharyngitis (4112343)<br>70 (70.0%) | Acute viral pharyngitis (4112343)<br>2403 (91.4%) | Acute viral pharyngitis (4112343)<br>1664 (63.3%) | Streptococcal sore throat (28060)<br>43 (82.7%) | Viral sinusitis (40481087)<br>17 (94.4%)        | Acute viral pharyngitis (4112343)<br>12 (66.7%) |
| 3   | Acute viral pharyngitis (4112343)<br>75 (76.5%) | Acute bronchitis (260139)<br>65 (65.0%)         | Acute bronchitis (260139)<br>2292 (87.2%)         | Acute bronchitis (260139)<br>1515 (57.6%)         | Otitis media (372328)<br>41 (78.8%)             | Otitis media (372328)<br>13 (72.2%)             | Acute cholecystitis (198809)<br>1 (5.6%)        |
| 4   | Otitis media (372328)<br>69 (70.4%)             | Osteoarthritis (80180)<br>58 (58.0%)            | Otitis media (372328)<br>1935 (73.6%)             | Chronic sinusitis (257012)<br>251 (9.5%)          | Sprain of ankle (81151)<br>35 (67.3%)           | Osteoarthritis (80180)<br>12 (66.7%)            | Alzheimer's disease (378419)<br>1 (5.6%)        |
| 5   | Streptococcal sore throat (28060)<br>59 (60.2%) | Angiodysplasia of stomach (4310024)<br>9 (9.0%) | Streptococcal sore throat (28060)<br>1399 (53.2%) | Cerebrovascular accident (381316)<br>194 (7.4%)   | Laceration of foot (4109685)<br>5 (9.6%)        | Streptococcal sore throat (28060)<br>9 (50.0%)  | Anemia (439777)<br>1 (5.6%)                     |

# tidy

You can create a tidy table that you can explore with the `tidy()` function:

```
tidy(result)
```

```
# A tibble: 125 × 10
```

|    | cdm_name | cohort_name | variable_name             | variable_level | concept_id | count | percentage | analysis | table_name   | type  |
|----|----------|-------------|---------------------------|----------------|------------|-------|------------|----------|--------------|-------|
|    | <chr>    | <chr>       | <chr>                     | <chr>          | <chr>      | <int> | <dbl>      | <chr>    | <chr>        | <chr> |
| 1  | GiBleed  | anemia      | Acute allergic reacti...  | -inf to -1     | 4084167    | 5     | 5.1        | standard | condition... | event |
| 2  | GiBleed  | anemia      | Acute bronchitis          | -inf to -1     | 260139     | 77    | 78.6       | standard | condition... | event |
| 3  | GiBleed  | anemia      | Acute viral pharyngit...  | -inf to -1     | 4112343    | 75    | 76.5       | standard | condition... | event |
| 4  | GiBleed  | anemia      | Angiodysplasia of sto...  | -inf to -1     | 4310024    | 7     | 7.14       | standard | condition... | event |
| 5  | GiBleed  | anemia      | Appendicitis              | -inf to -1     | 440448     | 6     | 6.12       | standard | condition... | event |
| 6  | GiBleed  | anemia      | Child attention defici... | -inf to -1     | 440086     | 6     | 6.12       | standard | condition... | event |
| 7  | GiBleed  | anemia      | Concussion with loss ...  | -inf to -1     | 375671     | 7     | 7.14       | standard | condition... | event |
| 8  | GiBleed  | anemia      | Cystitis                  | -inf to -1     | 195588     | 7     | 7.14       | standard | condition... | event |
| 9  | GiBleed  | anemia      | Facial laceration         | -inf to -1     | 4156265    | 9     | 9.18       | standard | condition... | event |
| 10 | GiBleed  | anemia      | First degree burn         | -inf to -1     | 4296204    | 8     | 8.16       | standard | condition... | event |

```
# i 115 more rows
```

# tidy

## Note

The `tidy()` function works for any `<summarised_result>` object that we create.

# summariseCohortOverlap

When creating multiple cohorts, we might be interested in the overlap of individuals between those cohorts. `summariseCohortOverlap()` does this for you.

```
result <- cdm$my_cohort |>
  summariseCohortOverlap()

glimpse(result)
```

Rows: 72

Columns: 13

```
$ result_id      <int> 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1...
$ cdm_name       <chr> "GiBleed", "GiBleed", "GiBleed", "GiBleed", "GiBleed", "GiBleed", "GiBleed", "GiBleed", "GiBleed", ...
$ group_name     <chr> "cohort_name_reference &&& cohort_name_comparator", "cohort_name_reference &&& cohort_na...
$ group_level    <chr> "anemia &&& osteoarthritis", "anemia &&& osteoarthritis", "anemia &&& osteoarthritis", "...
$ strata_name    <chr> "overall", "overall", "overall", "overall", "overall", "overall", "overall", "overall", "overall", ...
$ strata_level   <chr> "overall", "overall", "overall", "overall", "overall", "overall", "overall", "overall", "overall", ...
$ variable_name  <chr> "Only in reference cohort", "In both cohorts", "Only in comparator cohort", "Only in ref...
$ variable_level <chr> "Subjects", "Subjects", "Subjects", "Subjects", "Subjects", "Subjects", "Subjects", "Subjects", "Sub...
$ estimate_name  <chr> "count", "count", "count", "count", "count", "count", "count", "count", "count", "count", "count"...
$ estimate_type  <chr> "integer", "integer", "integer", "integer", "integer", "integer", "integer", "integer", "integer", ...
$ estimate_value <chr> "3", "97", "2533", "93", "7", "45", "99", "1", "17", "2533", "97", "3", "2578", "52", "0...
$ additional_name <chr> "overall", "overall", "overall", "overall", "overall", "overall", "overall", "overall", "overall", ...
```



# tableCohortOverlap

You can also get a table with `tableCohortOverlap()`.

```
tableCohortOverlap(result)
```

| Cohort name reference | Cohort name comparator | Estimate name | Variable name            |                 |                           |
|-----------------------|------------------------|---------------|--------------------------|-----------------|---------------------------|
|                       |                        |               | Only in reference cohort | In both cohorts | Only in comparator cohort |
| GiBleed               |                        |               |                          |                 |                           |
| anemia                | osteoarthritis         | N (%)         | 3 (0.11%)                | 97 (3.68%)      | 2,533 (96.20%)            |
|                       | pneumonia              | N (%)         | 93 (64.14%)              | 7 (4.83%)       | 45 (31.03%)               |
|                       | rheumatoid_arthritis   | N (%)         | 99 (84.62%)              | 1 (0.85%)       | 17 (14.53%)               |
| osteoarthritis        | pneumonia              | N (%)         | 2,578 (98.02%)           | 52 (1.98%)      | 0 (0.00%)                 |
|                       | rheumatoid_arthritis   | N (%)         | 2,612 (99.32%)           | 18 (0.68%)      | 0 (0.00%)                 |
| pneumonia             | rheumatoid_arthritis   | N (%)         | 52 (74.29%)              | 0 (0.00%)       | 18 (25.71%)               |

# summariseCohortTiming

When creating multiple cohorts, we might be interested in the timing of entry between cohorts. `summariseCohortTiming()` does this for you.

```
result <- cdm$my_cohort |>
  summariseCohortTiming(restrictToFirstEntry = TRUE)

glimpse(result)
```

Rows: 8,274

Columns: 13

```
$ result_id      <int> 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1...
$ cdm_name       <chr> "GiBleed", "GiBleed", "GiBleed", "GiBleed", "GiBleed", "GiBleed", "GiBleed", "GiBleed", "GiBleed", ...
$ group_name     <chr> "cohort_name_reference &&& cohort_name_comparator", "cohort_name_reference &&& cohort_na...
$ group_level    <chr> "anemia &&& osteoarthritis", "anemia &&& osteoarthritis", "anemia &&& osteoarthritis", "...
$ strata_name    <chr> "overall", "overall", "overall", "overall", "overall", "overall", "overall", "overall", "overall", ...
$ strata_level   <chr> "overall", "overall", "overall", "overall", "overall", "overall", "overall", "overall", "overall", ...
$ variable_name  <chr> "number records", "number subjects", "days_between_cohort_entries", "days_between_cohort...
$ variable_level <chr> NA, NA, NA, NA, NA, NA, NA, "density_001", "density_002", "density_003", "density_004", ...
$ estimate_name  <chr> "count", "count", "min", "q25", "median", "q75", "max", "density_x", "density_x", "densi...
$ estimate_type  <chr> "integer", "integer", "integer", "integer", "integer", "integer", "integer", "integer", "numeric", ...
$ estimate_value <chr> "97", "97", "-17715", "-4915", "2205", "6680", "15530", "-16313.4", "-16251.7980430528",...
$ additional_name <chr> "overall", "overall", "overall", "overall", "overall", "overall", "overall", "overall", "overall", ...
```



# tableCohortTiming

You can also get a table with `tableCohortTiming()`.

```
tableCohortTiming(result = result,
                  timeScale = "years",
                  uniqueCombinations = FALSE)
```

| Cohort name reference | Cohort name comparator | Variable name                | Estimate name      | Estimate value           |
|-----------------------|------------------------|------------------------------|--------------------|--------------------------|
| GiBleed               |                        |                              |                    |                          |
| anemia                | osteoarthritis         | number records               | N                  | 97                       |
|                       |                        | number subjects              | N                  | 97                       |
|                       |                        | years_between_cohort_entries | Median [Q25 - Q75] | 6.04 [-13.46 - 18.29]    |
|                       |                        |                              | Range              | -48.50 to 42.52          |
|                       | pneumonia              | number records               | N                  | 7                        |
|                       |                        | number subjects              | N                  | 7                        |
|                       |                        | years_between_cohort_entries | Median [Q25 - Q75] | 29.90 [15.57 - 38.99]    |
|                       |                        |                              | Range              | 1.77 to 53.45            |
|                       | rheumatoid_arthritis   | number records               | N                  | 1                        |
|                       |                        | number subjects              | N                  | 1                        |
|                       |                        | years_between_cohort_entries | Median [Q25 - Q75] | -17.99 [-17.99 - -17.99] |
|                       |                        |                              | Range              | -17.99 to -17.99         |
| osteoarthritis        | anemia                 | number records               | N                  | 97                       |
|                       |                        | number subjects              | N                  | 97                       |
|                       |                        | years_between_cohort_entries | Median [Q25 - Q75] | -6.04 [-18.29 - 13.46]   |
|                       |                        |                              | Range              | -42.52 to 48.50          |
|                       | pneumonia              | number records               | N                  | 52                       |

| Cohort name reference        | Cohort name comparator | Variable name                | Estimate name      | Estimate value           |
|------------------------------|------------------------|------------------------------|--------------------|--------------------------|
|                              |                        | number subjects              | N                  | 52                       |
|                              |                        | years_between_cohort_entries | Median [Q25 - Q75] | 43.49 [38.25 - 50.44]    |
|                              |                        |                              | Range              | 16.24 to 61.03           |
|                              | rheumatoid_arthritis   | number records               | N                  | 18                       |
|                              |                        | number subjects              | N                  | 18                       |
|                              |                        | years_between_cohort_entries | Median [Q25 - Q75] | 5.61 [-2.01 - 15.46]     |
|                              |                        |                              | Range              | -18.11 to 34.77          |
|                              | pneumonia              | anemia                       | number records     | N                        |
| number subjects              |                        |                              | N                  | 7                        |
| years_between_cohort_entries |                        |                              | Median [Q25 - Q75] | -29.90 [-38.99 - -15.57] |
|                              |                        |                              | Range              | -53.45 to -1.77          |
| osteoarthritis               |                        | number records               | N                  | 52                       |
|                              |                        | number subjects              | N                  | 52                       |
|                              |                        | years_between_cohort_entries | Median [Q25 - Q75] | -43.49 [-50.44 - -38.25] |
|                              |                        |                              | Range              | -61.03 to -16.24         |
| rheumatoid_arthritis         | anemia                 | number records               | N                  | 1                        |
|                              |                        | number subjects              | N                  | 1                        |
|                              |                        | years_between_cohort_entries | Median [Q25 - Q75] | 17.99 [17.99 - 17.99]    |
|                              |                        |                              | Range              | 17.99 to 17.99           |
|                              | osteoarthritis         | number records               | N                  | 18                       |
|                              |                        | number subjects              | N                  | 18                       |
|                              |                        | years_between_cohort_entries | Median [Q25 - Q75] | -5.61 [-15.46 - 2.01]    |
|                              |                        |                              | Range              | -34.77 to 18.11          |



# Your turn!

## Exercise 2 - large scale characteristics for anemia cohort

Can you get the large-scale characteristics from the **drug\_exposure** table for the **anemia** cohort table using an anytime-prior time window (excluding index date)? Visualise the result showing the top 5 most frequent codes.

►  [Click to see solution](#)

| Top | Estimate value                                                     |
|-----|--------------------------------------------------------------------|
| 1   | Acetaminophen 325 MG Oral Tablet (1127433)<br>79 (80.6%)           |
| 2   | Aspirin 81 MG Oral Tablet (19059056)<br>71 (72.5%)                 |
| 3   | poliovirus vaccine, inactivated (40213160)<br>64 (65.3%)           |
| 4   | Penicillin V Potassium 250 MG Oral Tablet (19133873)<br>54 (55.1%) |
| 5   | Ampicillin 100 MG/ML Injectable Solution (19129655)<br>9 (9.2%)    |

# CohortCharacteristics

- 👉 [Package website](#)
- 👉 [CRAN link](#)
- 👉 [Manual](#)

